SatsPath v2 Name Events

Goal

Make the authoritative server an admission and storage service, never the author of identity state. Every accepted mutation must be independently verifiable from the signed event history.

Event Model

Canonical events for the lifecycle of a namespace identifier:

- `Register`
- `ProfileUpdate`
- `RotateOwnerKey`
- `Revoke`
- `RecoverKey` (optional, disabled by default)

Every event MUST bind:

- `identifier_hash`
- Namespace identifier
- `action` (event type)
- `sequence`
- `previous_event_hash`
- `profile_hash`
- policy epoch
- validity/freshness fields (`created_at`)

Authorization Rules

1. **Registration**: Requires namespace issuance authorization plus proof that the proposed owner key accepts the binding (sequence 0, no previous event hash).
2. **Profile Updates**: Requires the signature of the current owner key.
3. **Key Rotation**: Requires authorization by the old key and acceptance by the new key.
4. **Revocation**: Requires the current owner key. This is a terminal state.
5. **Continuous Sequence**: Sequence must advance by exactly one.
6. **Chaining**: `previous_event_hash` must match the cryptographic hash of the current state.
7. **Compare-and-set**: Concurrent mutations use compare-and-set semantics; only one successor to a history head can commit.
8. **No Server Forgery**: A server must reject a syntactically valid self-signed replacement key that lacks history continuity.

Namespace Policy

Namespaces control admission, reserved names, and expiration, but they do NOT gain the ability to forge owner updates. Policy changes must be versioned and cannot retroactively weaken already-issued identifiers.